

Vlan Hopping – Switch Spoofing Attack

Warning:

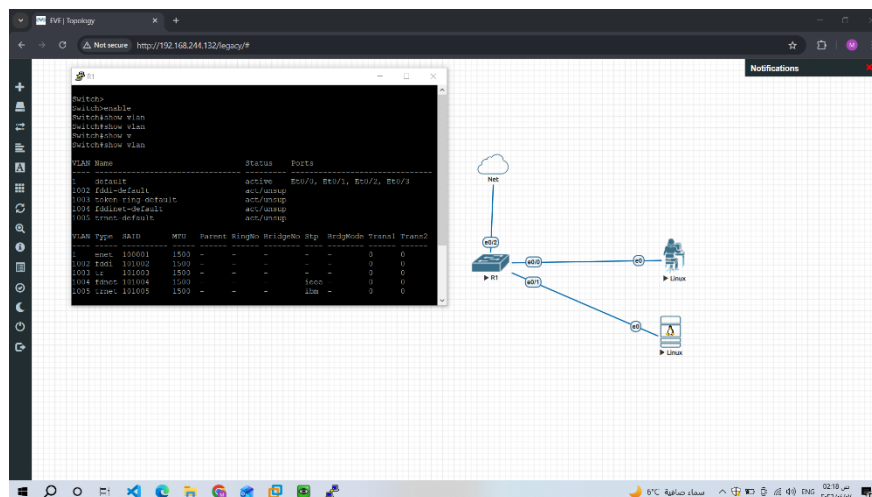
This environment was created in a controlled lab environment for educational purposes only.

1) General Information

- **Assessment Date:** 7/4/2026
- **Analyst Name:** Mustafa Jbili
- **Scope of Work:** Cybersecurity Analyst
- **Objective:** demonstrates how layer 2 applications (Vlans) can be vulnerable to threat acts by manipulate protocols such as DTP and how important is to have a appropriate configurations

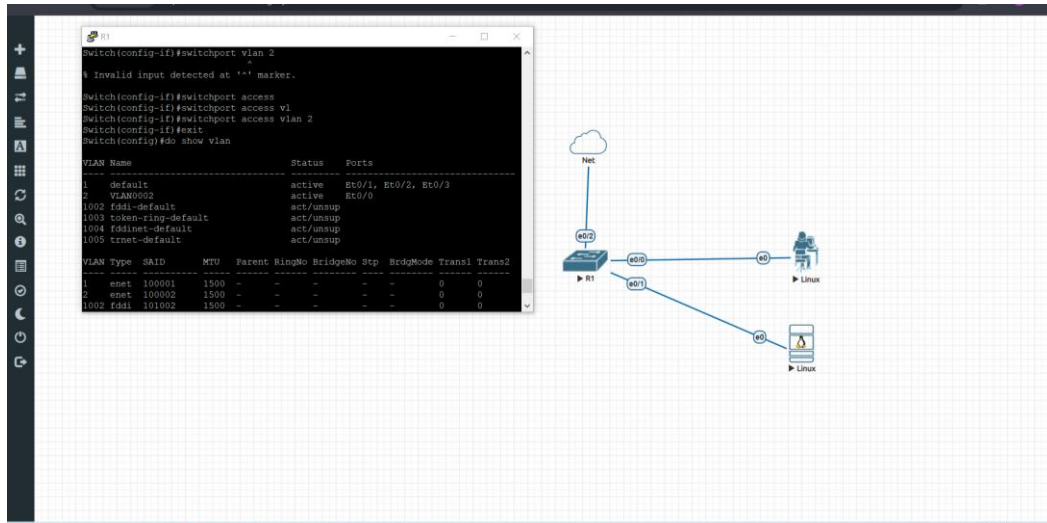
2) Lab's Environment :

- **Attacker Machine:** Kali Linux
- **Target Application:** layer 2 device (switch)
- **Deployment Method:** using tools such "Yersinia" which allows to send a Customized protocol

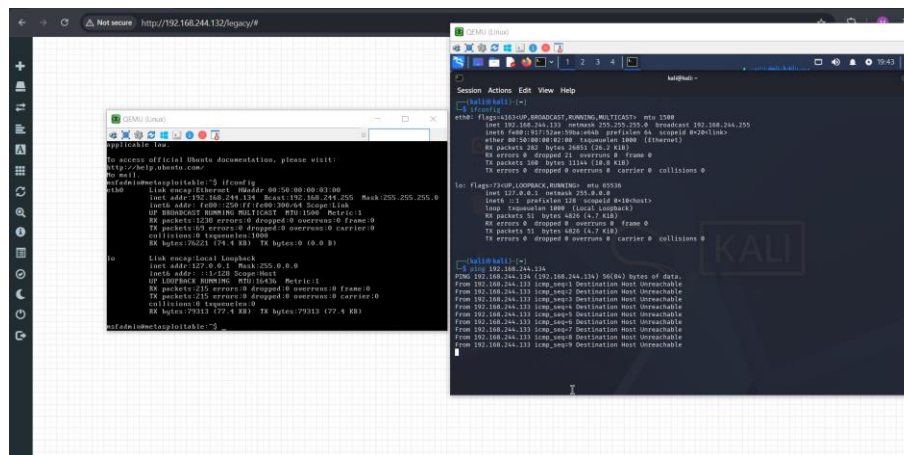


As shown in image above , conforming that all ports are on the same vlan (default one) .

- To do the task it required to move every device to different vlan to have the correct scenario for the attack



vlan 2 has been created and attacker environment was moved to vlan 2



- To confirm that the attacker environment was moved successfully a ping test was applied to clarify that

3) yersinia :

- Using Yersinia, the attacker sends spoofed DTP packets to the switch, pretending to be another switch. As a result, the target switch accepts the negotiation request and converts the port into **trunk mode**.

4) Attack Execution:

- The attacker machine (Kali Linux) is connected to a switch port configured in dynamic mode.
- The attacker launches Yersinia and sends DTP negotiation frames.
- The switch receives the forged packets and assumes it is communicating with a legitimate switch.
- The switch port transitions from access mode to trunk mode.

4) Impact of the Attack:

Before the attack:

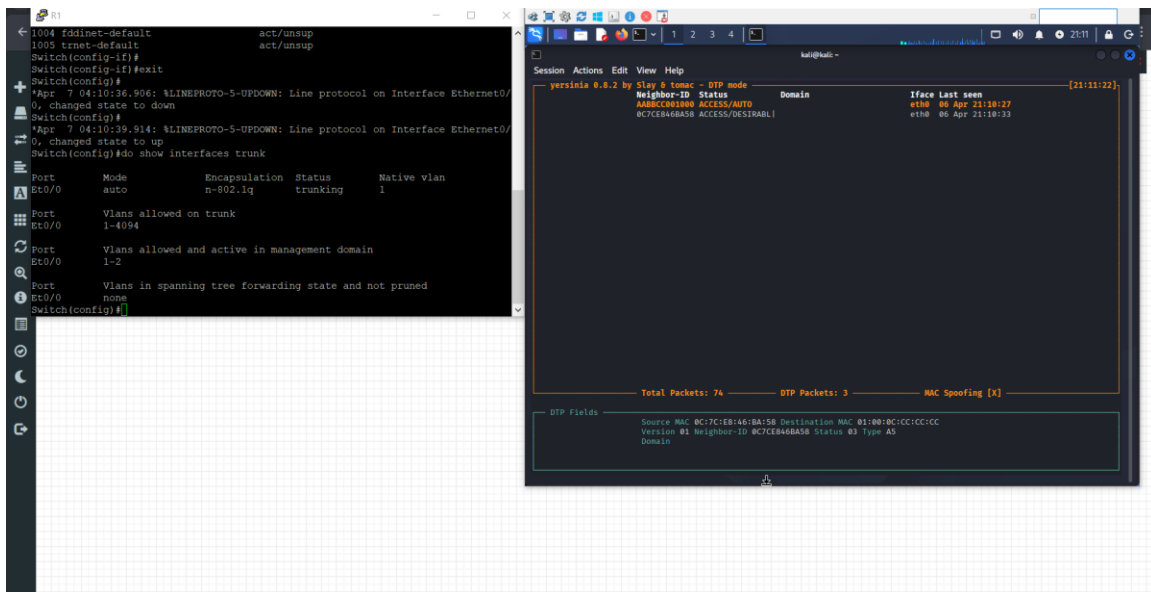
- The attacker is restricted to a single VLAN.
- No communication with other VLANs is possible.

After the attack:

- The attacker gains access to multiple VLANs through the trunk link.
- VLAN segmentation is effectively bypassed.
- The attacker can communicate with devices in other VLANs.

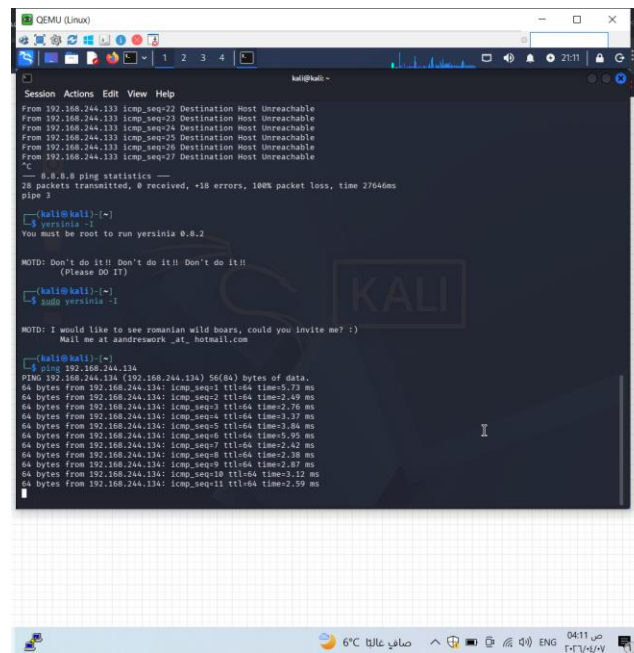
This demonstrates a critical breakdown of network isolation and segmentation.

5) Select security level:



- by using yersinia , spoofing was made with successfully and port was moved to trunking mode by that packets will be transmitted by default to kali device

6) testing the lab:



- Kali managed to ping metasploitable even it was on different vlan